Mohamed Raiyan

SOFTWARE ENGINEER | AI ENGINEER

Eskilstuna, Sweden | +46767516351 | raiyanmfm@gmail.com | https://mohamedraiyans.github.io/

Professional Summary

Software and AI engineer with 9+ years building scalable backend systems, high-performance APIs and AI-powered applications. Current work centres on LLM and SLM integration: retrieval-augmented generation, local and quantised model deployment, and multi-provider LLM architecture in Python and TypeScript. MSc in Computer and Systems Sciences, Stockholm University, with a thesis on generative AI and multimodal interaction.

Technical Skills

Languages	Python, TypeScript, JavaScript, PHP, Java
Frameworks	FastAPI, Django, NestJS, Node.js, React, Next.js, Laravel
Databases	PostgreSQL and MySQL (2TB+ scale), Chroma (vector), NoSQL, Airtable
APIs & Services	REST, Shopify, eBay, AWS integrations
AI / LLM	Azure OpenAI, OpenAI and Claude APIs, LangChain, retrieval-augmented generation (RAG), vector databases (ChromaDB), local and quantised model deployment, prompt engineering, multimodal AI
ML & Data Science	scikit-learn, pandas, NumPy, Classification, Clustering
Cloud & DevOps	Azure, Docker, CI/CD, Agile/Scrum

Experience

Freelance Software Engineer / AI Engineer (Remote) **Apr 2026 – Present**

- Delivering remote client engagements alongside independent AI engineering projects, working end to end from architecture through deployment.
- Knowledge assistant (TypeScript, NestJS, Next.js, Chroma, Redis, PostgreSQL): retrieval-augmented generation with structure-aware chunking and per-document chunk caps for retrieval diversity, built on a LangChain retriever with an LCEL generation chain. Three model providers switchable at runtime, including an open-weight model alongside hosted proprietary ones, with embeddings computed locally. AES-256-GCM encryption of provider credentials, version-counter cache invalidation, and provider rate-limit header tracking.
- Speech recognition service (Python, FastAPI, faster-whisper on CTranslate2): small, specialized model run locally on CPU with int8 quantisation. Decoding strategy, thread count and audio segmentation chosen by measurement rather than by library default, giving roughly 2x faster decoding at matching accuracy and a further 2.2x from correct thread configuration. Model identifier and revision recorded on every response; background processing via a job queue with retry, backoff and permanent-failure handling.
- Document analysis application (Python, FastAPI, React, Azure OpenAI and Azure AI Document Intelligence): users upload a document and receive model-generated assessment through a designed interface, with JWT and Google Sign-In authentication, containerised with Docker Compose.

- Multi-tenant SaaS backend (FastAPI, PostgreSQL with Alembic, Celery and Redis): shared-database tenant isolation, background job processing, and AI-assisted listing enrichment.

Imara Software Solutions | Senior Software Engineer / AI Engineer **Jan 2019 – Mar 2026**

- Designed and maintained scalable backend systems handling over 500,000 API requests daily across distributed cloud infrastructure.
- Developed and integrated AI-powered applications using large and small language models, including OpenAI and Claude APIs, LangChain, retrieval-augmented generation and prompt engineering, for automation and intelligent data processing.
- Built and maintained RESTful APIs and backend services in Python (Django, FastAPI) and Node.js, with React-based admin panels and internal tools.
- Containerised microservices with Docker and maintained CI/CD pipelines for automated testing and deployment across staging and production.
- Managed high-volume PostgreSQL and MySQL databases (2TB+) with automated backups and optimised indexing, and implemented monitoring dashboards for API uptime, service health and error rates.
- Led integration of third-party APIs including Shopify, eBay and AWS services; mentored junior developers and led sprint planning.

Imara Software Solutions | Software Developer **Nov 2014 – Dec 2018**

- Built custom CMS solutions, dynamic web applications and e-commerce platforms using PHP (Laravel, CodeIgniter) and JavaScript, with REST APIs and admin panel development.
- Managed MySQL databases including schema migrations and cross-site data syncing, with versioned deployments via Git-based CI pipelines.

Education

Stockholm University, Master's degree, Computer and Systems Sciences **Sep 2023 - Oct 2025**

- **Major**: Machine Learning, IT Management, Research Methodology, ERP Systems, AI, Internet of Things Services, Information Security, Data Mining, Cyber Forensics
- **Thesis: Programming through Sketching – Non-Linguistic Interaction with Generative AI.** Explored sketch-based UI generation using generative AI via multimodal inputs.

University of Greenwich, Bachelor's Degree, Information Technology **2013 - 2014**

Major: Web Engineering, Database Engineering, Programming Frameworks

Publications

Programming Through Sketching: Non-Linguistic Interaction with Generative AI. Independent thesis, advanced level (Master, two years), 30 HE credits, Department of Computer and Systems Sciences, Stockholm University, 2025.

- https://www.diva-portal.org/smash/record.jsf?pid=diva2:1999524